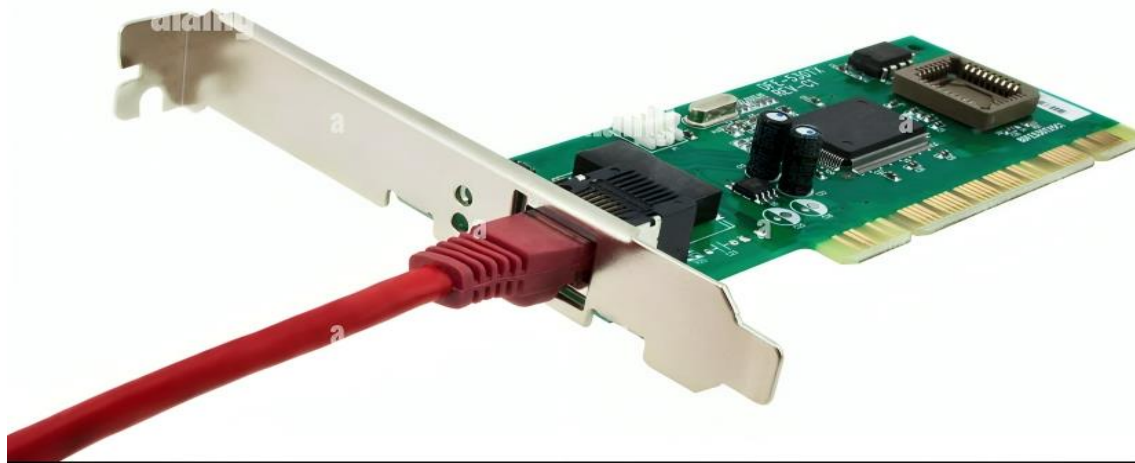


Network Hacking

Network Sniffing

Before we start Network Hacking we need to know
Network interface card (NIC)



Network Sniffing



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It can be on two modes

First (Managed Mode) it is normal mode to take internet and don't make hostspot

Monitor Mode (hacking mode)

It is for capturing the data and making hostspot

When we work we will convert NIC to monitor mode

Network Sniffing



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Sniffing is to capture packets of data transmitted by other computers

Having the packets will allow us to read and analyze sensitive data , password, visa card , usernames etc...

Network Sniffing



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The most popular way to attack network is man in the middle attack or (MITM)



Network Protocols



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When You sniff if found http not https then it would be easy to capture the data and analyze it as

`http://testphp.vulnweb.com/`

But if you found https then it is secure and you need to downgrade it to http

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To downgrade the https to http you will use (sslstrip tool) in linux

After using sslstrip to downgrade https to http then use ettercap tool with the network interface card in (monitor mode)

Wireshark



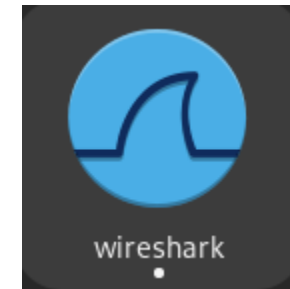
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The next step is to use wireshark to analyze the data of packets

It shows you all packets going from victim

To server and vice verse

It is build in linux system just write
writeshark

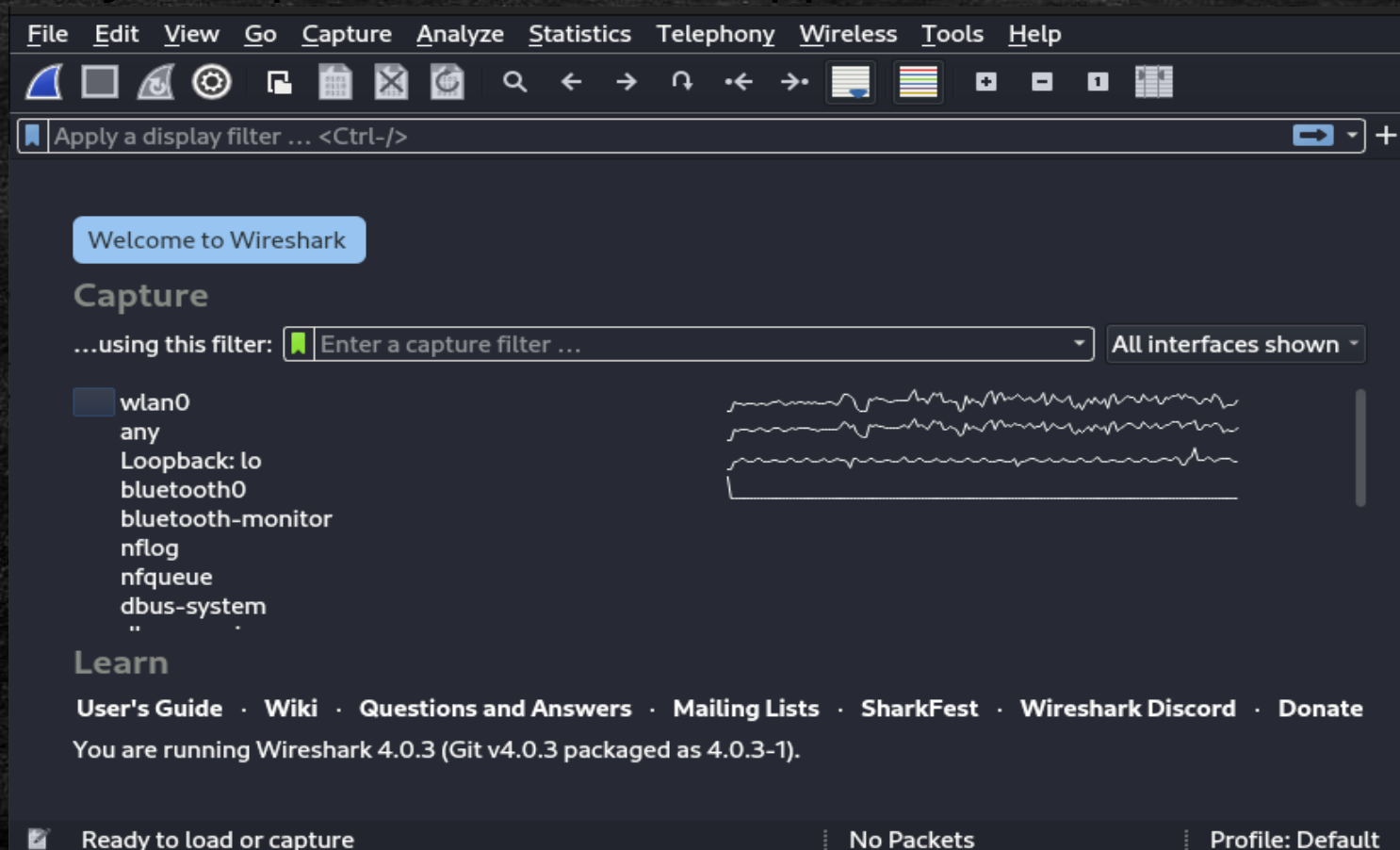


Wireshark



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- Once you opened it will appear like that



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Choose wlan0 or wlan0mon if found

After entering you will find thousands of packets every minute

Observe the titles (source , destination , protocol)

You need to focus on protocol and search for udp or ftp or ssh or

The Red Nexus logo, featuring a stylized orange and red hexagonal icon to the left of the text "RED nexus" in a white, sans-serif font. The word "RED" is in all caps, and "nexus" is in lowercase with a red diagonal line through the 'x'.

Wireshark

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
7491	45.0972963...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9207825 Win=1
7492	45.1027157...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9209205 Win=1
7493	45.1037517...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9210585 Win=1
7494	45.1077175...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9211965 Win=1
7495	45.1125304...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9213345 Win=1
7496	45.1149466...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9214725 Win=1
7497	45.1244781...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9216105 Win=1
7498	45.1246754...	52.108.24.6	192.168.1.6	TCP	60	443 → 40278 [ACK] Seq=47906 Ack=9217485 Win=1

Frame 1: 13854 bytes on wire (110832 bits), 13854 bytes captured on interface eth0, capture length 13854 bytes

- Ethernet II, Src: b8:9a:2a:63:02:63 (b8:9a:2a:63:02:63), Dst: b8:9a:2a:63:02:63 (b8:9a:2a:63:02:63)
- Internet Protocol Version 4, Src: 192.168.1.6, Dst: 52.108.24.6
- Transmission Control Protocol, Src Port: 40278, Dst Port: 443
- Transport Layer Security**

```

0000 d8 e8 44 b8 bf 04 b8 9a 2a 63 02 63 08 00
0010 36 10 11 59 40 00 40 06 e5 6e c0 a8 01 00
0020 18 06 9d 56 01 bb 8c 85 10 ec 83 e5 3f 60
0030 20 11 44 23 00 00 5d f1 f7 35 8c a3 74 f0
0040 42 d4 a2 ef 22 59 71 7c d6 5a 74 f3 80 40
0050 8f 2c 55 68 6c ab 47 09 74 fa 1e 5d 97 e0
0060 25 90 3d ed 86 1d 46 f8 33 0b 52 de 3f 80
0070 65 cf d6 8e dd e0 1b 13 f5 c6 f4 7b d6 00
0080 f1 8a fb 0a fd 0a 9e 34 8e f0 64 49 f5 10
0090 0d 0f 2b f7 f2 81 bc 26 db 42 f5 b4 05 20
    
```

Network Sniffing



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File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
331...	150.151371...	192.168.1.6	44.228.249.3	HTTP	499	GET / HTTP/1.1
331...	150.430446...	44.228.249.3	192.168.1.6	HTTP	1297	HTTP/1.1 200 OK (text/html)
331...	152.479100...	192.168.1.6	52.219.56.68	HTTP	407	GET /aiscripts/scripts.json?_=1719476435461 H
331...	153.087684...	52.219.56.68	192.168.1.6	HTTP	450	HTTP/1.1 200 OK

Frame 33103: 499 bytes on wire (3992 bits), 499 bytes captured on interface 0

Ethernet II, Src: b8:9a:2a:63:02:63 (b8:9a:2a:63:02:63), Dst: 08:00:27:00:00:00

Internet Protocol Version 4, Src: 192.168.1.6, Dst: 44.228.249.3

Transmission Control Protocol, Src Port: 59932, Dst Port: 80

Hypertext Transfer Protocol

0000 d8 e8 44 b8 bf 04 b8 9a 2a 63 02 63 08 00
0010 01 e5 f9 02 40 00 40 06 58 7a c0 a8 01 00
0020 f9 03 ea 1c 00 50 a9 b4 23 1d b4 11 65 d0
0030 00 fb e9 6d 00 00 01 01 08 0a 8f be 12 3f
0040 f5 51 47 45 54 20 2f 20 48 54 54 50 2f 3f
0050 0d 0a 48 6f 73 74 3a 20 74 65 73 74 70 6f
0060 76 75 6c 6e 77 65 62 2e 63 6f 6d 0d 0a 4d
0070 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2f
0080 69 76 65 0d 0a 55 70 67 72 61 64 65 2d 4d
0090 65 63 75 72 65 2d 52 65 71 75 65 73 74 7f

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Then I need you to visit

- <http://testphp.vulnweb.com/login.php>

And sign in with username: test and password: test

Go back to wireshark and observe there is POST request with the sensitive data

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805...	348.124846...	44.228.249.3	192.168.1.6	HTTP	158 HTTP/1.1 200 OK (text/html)
806...	349.617745...	192.168.1.6	52.219.144.56	HTTP	407 GET /aiscripts/scripts.json?_=17194766326
806...	350.132485...	52.219.144.56	192.168.1.6	HTTP	450 HTTP/1.1 200 OK
806...	352.218331...	192.168.1.6	44.228.249.3	HTTP	710 POST /userinfo.php HTTP/1.1 (application)
807...	352.438880...	44.228.249.3	192.168.1.6	HTTP	294 HTTP/1.1 200 OK (text/html)
818...	422.630379...	192.168.1.6	44.228.249.3	HTTP	586 GET /logout.php HTTP/1.1
819...	422.846038...	44.228.249.3	192.168.1.6	HTTP	2644 HTTP/1.1 200 OK (text/html)
820...	424.102741...	192.168.1.6	52.219.144.56	HTTP	407 GET /aiscripts/scripts.json?_=17194767070
820...	424.420458...	52.219.144.56	192.168.1.6	HTTP	450 HTTP/1.1 200 OK
820...	425.556176...	192.168.1.6	44.228.249.3	HTTP	556 GET /login.php HTTP/1.1
820...	425.773103...	44.228.249.3	192.168.1.6	HTTP	158 HTTP/1.1 200 OK (text/html)

▶ Frame 80682: 710 bytes on wire (5680 bits), 710 bytes captured on interface (5680 bits) on 0:8:9a:2a:63:02:63	01f0	6d 61 67 65 2f 77 65 62 70 2c 69 6d
▶ Ethernet II, Src: b8:9a:2a:63:02:63 (b8:9a:2a:63:02:63), Dst: b8:9a:2a:63:02:63	0200	61 70 6e 67 2c 2a 2f 2a 3b 71 3d 30
▶ Internet Protocol Version 4, Src: 192.168.1.6, Dst: 44.228.249.3	0210	70 70 6c 69 63 61 74 69 6f 6e 2f 73
▶ Transmission Control Protocol, Src Port: 48556, Dst Port: 80	0220	64 2d 65 78 63 68 61 6e 67 65 3b 76
▶ Hypertext Transfer Protocol	0230	71 3d 30 2e 37 0d 0a 52 65 66 65 72
▼ HTML Form URL Encoded: application/x-www-form-urlencoded	0240	68 74 74 70 3a 2f 2f 74 65 73 74 70
▶ Form item: "uname" = "test"	0250	75 6c 6e 77 65 62 2e 63 6f 6d 2f 6c
▶ Form item: "pass" = "test"	0260	2e 70 68 70 0d 0a 41 63 63 65 70 74
	0270	6f 64 69 6e 67 3a 20 67 7a 69 70 2c
	0280	6c 61 74 65 0d 0a 41 63 63 65 70 74
	0290	67 75 61 67 65 3a 20 65 6e 2d 55 53
	02a0	71 3d 30 2e 39 2c 61 72 3b 71 3d 30

Questions



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